

Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d -step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1 - \lambda_c) \delta \mathbf{x}[\mathbf{k}] - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k]f_T[k]$$

d -step measurement delay:

$$\delta \mathbf{x}[\mathbf{k}] = \delta x[k-d] + \sum_{i=1}^d \Delta x_d[k-i] - \sum_{i=1}^d a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta x_D[k-i]$$

Ideal control effort (with d -step measurement delay):

$$\begin{aligned} \bar{\bar{f}}_{dx}[k] = a_x^{-1}[k] \Big\{ & \underbrace{\left[\Delta x_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta x_d[k-i] \right]}_{\Delta x_d^d[k]} + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] \\ & - \underbrace{\left[\delta x_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta x_D[k-i] \right]}_{\delta x_D^d[k]} \Big\} \end{aligned}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\left\{ a_x[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_x[k-i]f_T[k-i] \right\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k] [f_{dx}[k] - \bar{\bar{f}}_{dx}[k]] - \delta x_T^d[k]$$

$$a_x[k] \bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] - \delta x_D^d[k]$$

$$a_x[k] f_{dx}[k] = (1 + e_{a_x}[k] \hat{a}_x^{-1}[k]) \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[(\delta x[k-d] + n_x[k]) - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &= e_{a_x}[k] f_{dx}[k] + (1 - \lambda_c) n_x[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_x[k-i] - \hat{a}_x[k-i]) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a'_x[k] x_{ram}[k]$$

$$\begin{aligned} a_x[k+1] &= \mathbf{a}_{x_{det}}[\mathbf{k} + \mathbf{1}] + a_{x_{ram}}[k+1] \\ &= \underbrace{\left\{ \mathbf{a}_{x_{det}}[\mathbf{k}] + a_{x_{ram}}[k] \right\}}_{a_x[k]} + \Delta \mathbf{a}_x[\mathbf{k}] + \underbrace{\left\{ a_{x_{ram}}[k+1] - a_{x_{ram}}[k] \right\}}_{\Delta a_{ram}[k]} \end{aligned}$$

$$a_x[k+1] = a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k]$$

$$\Delta a_x[k] = a_{x_{det}}[k+1] - a_{x_{det}}[k] = a'_x[k] \Delta h_d[k], \quad \Delta a_{ram}[k] = a'_x[k] \Delta x_{ram}[k]$$

$$\Delta h_d[k] = h_d[k+1] - h_d[k] \text{ (desired-trajectory height increment).}$$

$$\text{Parameter estimator:} \quad \hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k]$$

$$\text{Parameter estimation error:} \quad e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$\text{Estimation error dynamics:} \quad e_{a_x}[k+1] = e_{a_x}[k] + \Delta a_{ram}[k]$$

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - \sum_{j=1}^i \Delta a_{ram}[k-j]$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &\approx e_{a_x}[k] \underbrace{\left\{ f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ &\quad - \underbrace{\left\{ (1 - \lambda_c) \sum_{i=1}^d (d+1-i) f_{dx}[k-i] \Delta a_{ram}[k-i] \right\}}_{\delta x_{\delta af}^d[k]} \\ &\quad + \delta x_D^d[k] + (1 - \lambda_c) n_x[k] \end{aligned}$$

$$\boxed{\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \delta x_D^d[k] - \underbrace{\left\{ \delta x_T^d[k] - \delta x_{\delta af}^d[k] + (1 - \lambda_c) n_x[k] \right\}}_{\varepsilon[k]}}$$

State equation

($\delta x_D^d \equiv 0$: disturbance not modelled)

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k]$$

$$a_x[k+1] = a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k]$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^d \Delta a_x[k-i] \end{bmatrix} \text{ (known)}, \quad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k]$$

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta \hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta \hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \left(\hat{a}_x[k] - \sum_{i=1}^d \Delta a_x[k-i] \right) \\ &= \left(a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + r_2[k] \right) - \hat{a}_x[k] + \sum_{i=1}^d \Delta a_x[k-i] \\ &= e_{a_x}[k] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{aligned}$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k] e_{y_1}[k] + \ell_{12}[k] e_{y_2}[k]$$

$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k] e_{y_1}[k] + \ell_{22}[k] e_{y_2}[k]$$

$$\delta \hat{x}_3[k+1] = \lambda_c \cdot \delta \hat{x}_3[k] + \ell_{31}[k] e_{y_1}[k] + \ell_{32}[k] e_{y_2}[k]$$

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

Error dynamics

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k] - \ell_{11}[k] e_{\delta x_1}[k] - \ell_{12}[k] e_{a_x}[k]$$

$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k] - \ell_{21}[k] e_{\delta x_1}[k] - \ell_{22}[k] e_{a_x}[k]$$

$$e_{\delta x_3}[k+1] = \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] - \ell_{31}[k] e_{\delta x_1}[k] - \ell_{32}[k] e_{a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]}$$

$$e_{a_x}[k+1] = e_{a_x}[k] - \ell_{41}[k] e_{\delta x_1}[k] - \ell_{42}[k] e_{a_x}[k] + \underbrace{\Delta a_{ram}[k]}_{q_4[k]}$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k - i]$$

$$\mathbf{e}[k + 1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k] \mathbf{r}[k]$$

Gain fluctuation variance $\text{Var}(\Delta a_{ram})$

Assumption ($x = x_{det} + x_{ram}$, perfect tracking $x_{det} = x_d$): $\delta x = x_d - x = -x_{ram}$ (zero-mean), $x_{ram} = -\delta x$.

$$a'_x := \frac{da_x}{dh} = -\frac{a_x K_h}{R}, \quad K_h = \frac{1}{c} \frac{dc}{d\bar{h}} \text{ (wall sensitivity)}, \quad a_{x_{ram}} = a'_x x_{ram}, \quad x_{ram} = -\delta x$$

Closed-loop tracking variance (thermal-driven, $\hat{a}_x = a_x$):

$$\text{Var}(x_{ram}) = \text{Var}(\delta x) = C_{\delta x} \sigma_{\delta h}^2, \quad C_{\delta x} = 2 + \frac{1}{1 - \lambda_c^2}, \quad \sigma_{\delta h}^2 := 4k_B T a_x$$

Increment of a stationary signal ($\rho_1 = \text{Cov}(x_{ram}[k+1], x_{ram}[k]) / \text{Var}(x_{ram})$, lag-1 autocorr):

$$\text{Var}(\Delta x_{ram}) = 2 \text{Var}(x_{ram}) - 2 \text{Cov}(x_{ram}[k+1], x_{ram}[k]) = 2(1 - \rho_1) \text{Var}(x_{ram})$$

$$\rho_1 = \lambda_c + \frac{2(1 - \lambda_c)}{C_{\delta x}} \Rightarrow 1 - \rho_1 = \frac{1}{(1 + \lambda_c) C_{\delta x}}$$

Gain increment $\Delta a_{ram} = a'_x \Delta x_{ram}$ ($C_{\delta x}$ and $(1 - \rho_1)$ cancel):

$$\text{Var}(\Delta a_{ram}) = a_x'^2 \text{Var}(\Delta x_{ram}) = a_x'^2 \cdot \underbrace{2(1 - \rho_1) C_{\delta x}}_{= 2/(1 + \lambda_c)} \sigma_{\delta h}^2$$

$$\text{Var}(\Delta a_{ram}) = \frac{2}{1 + \lambda_c} a_x'^2 \sigma_{\delta h}^2$$

Impl.: K_h , $\sigma_{\delta h}^2 = 4k_B T a_\perp$ evaluated at $\bar{h}_{\text{meas}}$ (measured, p_m); gain $a_x \rightarrow \hat{a}_x$ (estimate).

Correction — $a_{x_{ram}}$ is first-order autoregressive (AR(1)), not a random walk

True-signal dynamics ($x_{ram} = -\delta x$; $\delta x[k+1] = \lambda_c \delta x[k] - F_{dx}[k] e_{a_x}[k] - \varepsilon[k]$):

$$a_{x_{ram}}[k+1] = -a'_x \delta x[k+1] = \lambda_c a_{x_{ram}}[k] + a'_x F_{dx}[k] e_{a_x}[k] + a'_x \varepsilon[k]$$

$$\Delta a_{ram}[k] = (\lambda_c - 1) a_{x_{ram}}[k] + a'_x F_{dx}[k] e_{a_x}[k] + a'_x \varepsilon[k]$$

True gain state ($a_x = a_{x_{det}} + a_{x_{ram}}$, $a_{x_{det}}[k+1] = a_{x_{det}}[k] + \Delta a_x[k]$, $e_{a_x} = a_x - \hat{a}_x$):

$$\begin{aligned} a_x[k+1] &= (a_{x_{det}}[k] + \Delta a_x[k]) + \lambda_c (a_x[k] - a_{x_{det}}[k]) + a'_x F_{dx}[k] (a_x[k] - \hat{a}_x[k]) + a'_x \varepsilon[k] \\ &= \underbrace{(\lambda_c + a'_x F_{dx}[k]) a_x[k]}_{\mathbf{F}_e(4,4)} + \underbrace{(1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k] - a'_x F_{dx}[k] \hat{a}_x[k]}_{\text{known input}} + \underbrace{a'_x \varepsilon[k]}_{q_4[k]} \end{aligned}$$

$$\boxed{\mathbf{F}_e(4,4) = \lambda_c + a'_x F_{dx}[k]}$$

Estimator ($(\lambda_c + a'_x F_{dx}) \hat{a}_x - a'_x F_{dx} \hat{a}_x = \lambda_c \hat{a}_x$: the feedthrough vanishes at the estimate):

$$\hat{a}_x[k+1] = \lambda_c \hat{a}_x[k] + (1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k] + \ell_{4\cdot}[k] e_y[k]$$

$$e_{a_x}[k+1] = a_x[k+1] - \hat{a}_x[k+1] = (\lambda_c + a'_x F_{dx}[k]) e_{a_x}[k] + a'_x \varepsilon[k] - \ell_{4\cdot}[k] e_y[k]$$

Process noise (white driver $w_4 = a'_x \varepsilon$; $\text{Var}(\varepsilon) = (1 - \lambda_c^2) \text{Var}(\delta x)$ since δx is AR(1)):

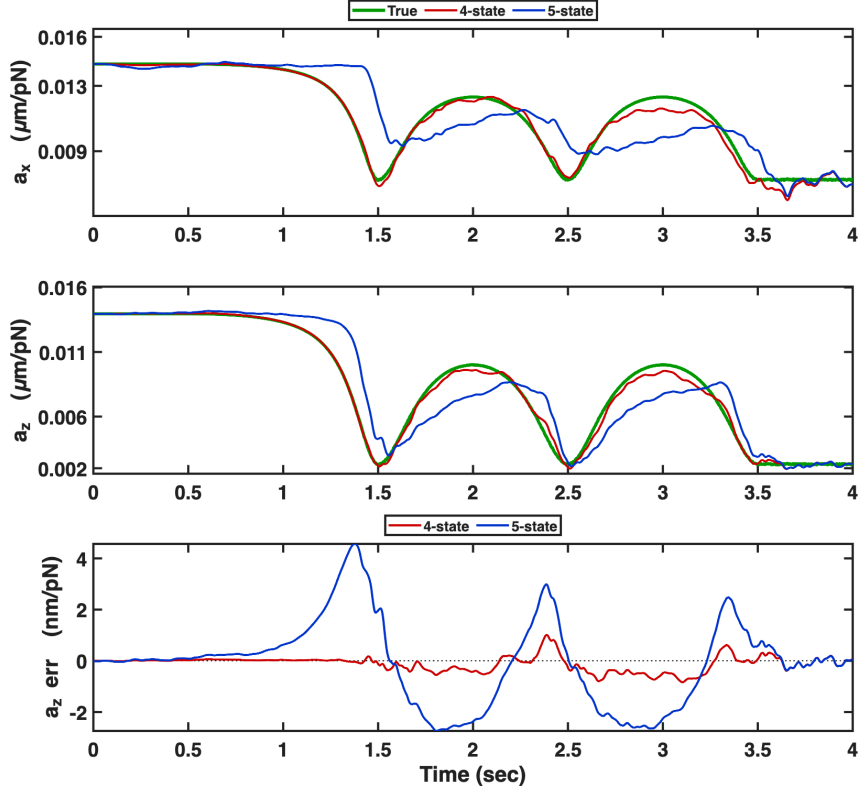
$$q_4 = \text{Var}(w_4) = a_x'^2 \text{Var}(\varepsilon) = a_x'^2 (1 - \lambda_c^2) \text{Var}(\delta x) = (1 - \lambda_c^2) C_{\delta x} a_x'^2 \sigma_{\delta h}^2$$

$$\boxed{q_4 = a_x'^2 \text{Var}(\varepsilon) = (3 - 2\lambda_c^2) a_x'^2 \sigma_{\delta h}^2}$$

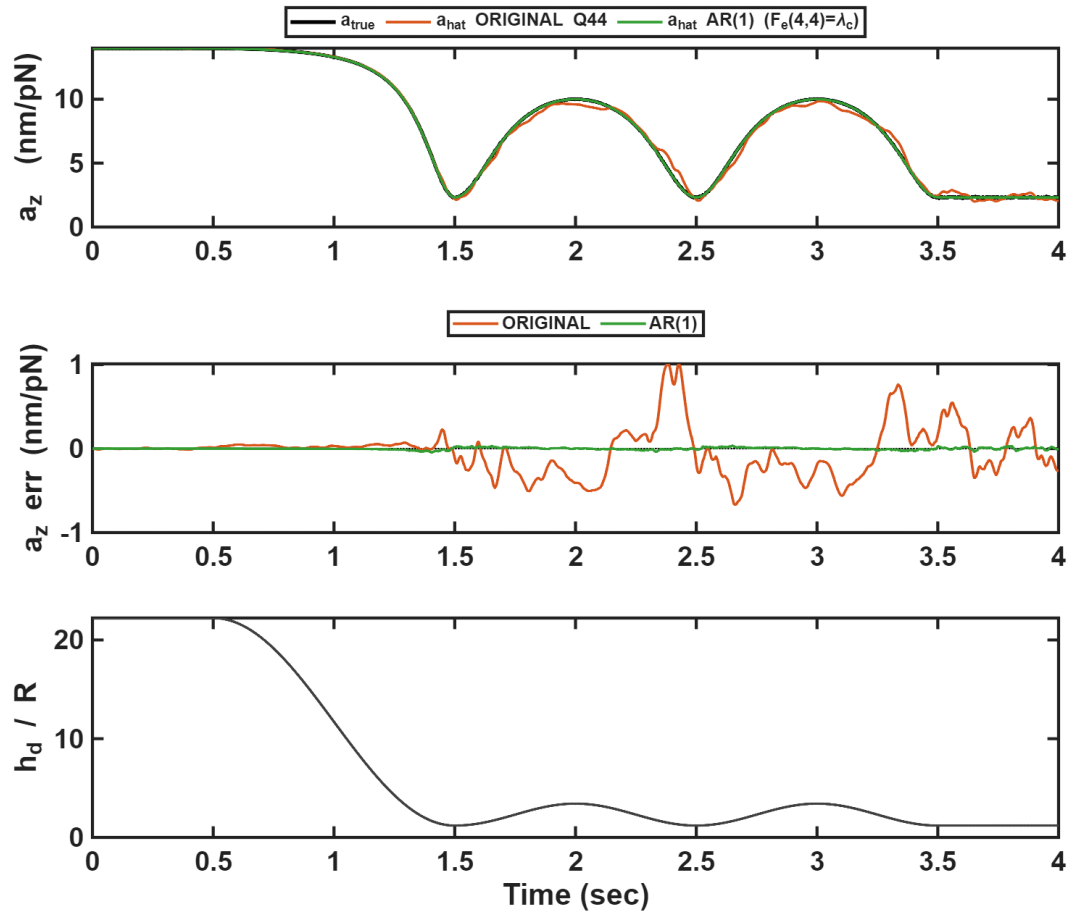
Impl.: predict and q_4 use $\hat{a}_x$ with a'_x at $\bar{h}_{\text{meas}}(p_m)$; reversion target $a_{x_{det}}$ from p_d ; production $\mathbf{F}_e(4,4) = \lambda_c$ (drops $a'_x F_{dx}$).

Verification (1 Hz, near wall)

Per-axis EKF, seed ensemble; near-wall 1 Hz oscillation ($h_{init} = 50 \mu\text{m}$, trough $\bar{h} = 1.2$).



a_x, a_z : true gain and the 4-state / 5-state estimates; bottom tile, the a_z estimation error $\hat{a}_x - a_x$.



a_z : original random-walk Q_{44} ($F_e(4,4) = 1$) vs $AR(1)$ reverting gain ($F_e(4,4) = \lambda_c$); middle tile, estimation error; bottom, $\bar{h}$. Near-wall $|\hat{a}_z - a_z|$: 9.4% $\rightarrow$ 0.4%.

Model-free a' via self-difference — original (RW) vs AR(1)

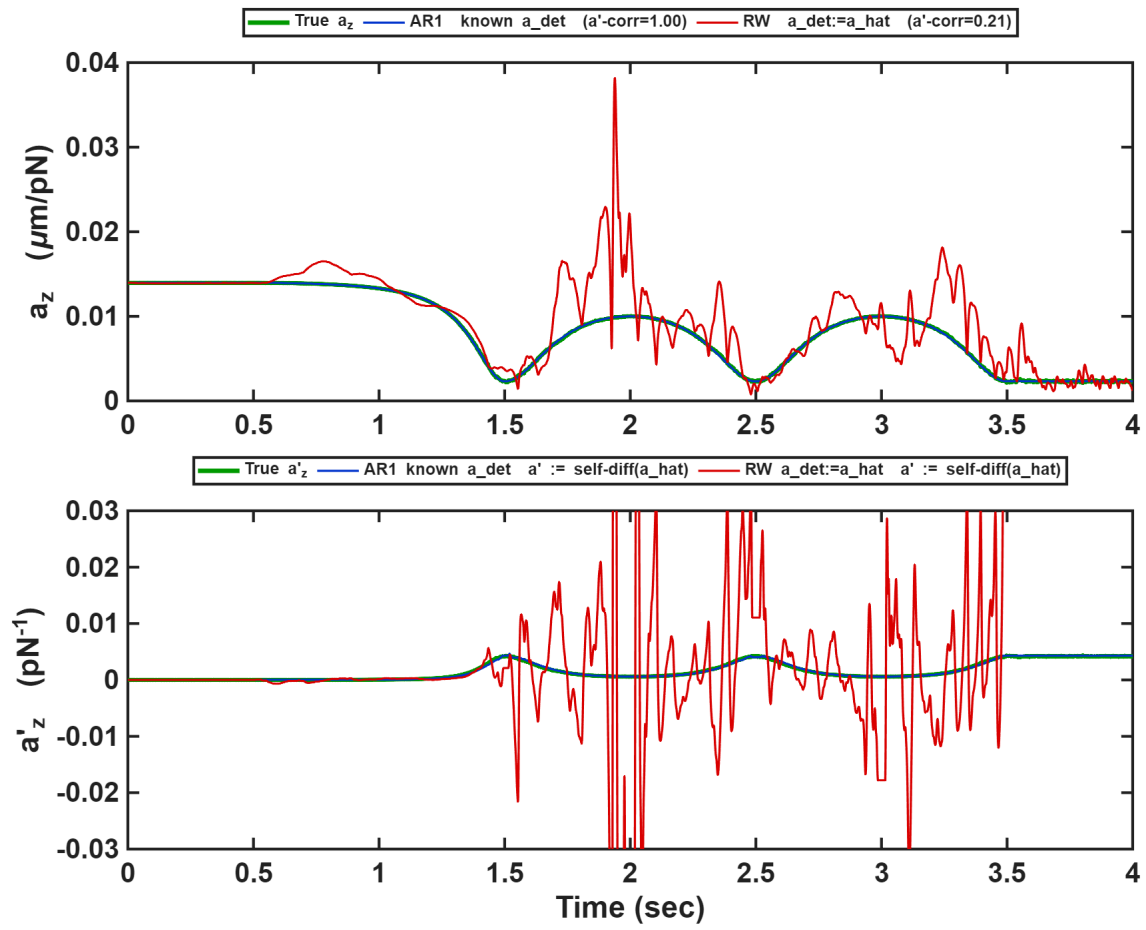
So far a'_x has come from the known wall model. If the wall is unknown, we can read a'_x straight off the gain estimate instead: take the change in $\hat{a}_x$ and divide it by the (known) change in desired height. But that raw ratio is very jumpy — $\hat{a}_x$ already carries estimation noise, and taking a difference makes the noise worse. So we do two simple things: (i) smooth the raw slope with a running average (an EWMA with weight β), and (ii) stop updating whenever the height is barely moving (a gate γ , so we never divide by almost zero):

$$\hat{a}'_x[k] = \begin{cases} (1 - \beta) \hat{a}'_x[k-1] + \beta \frac{\hat{a}_x[k] - \hat{a}_x[k-1]}{\Delta h_d[k-1]}, & |\Delta h_d[k-1]| > \gamma \\ \hat{a}'_x[k-1], & |\Delta h_d[k-1]| \leq \gamma \quad (\text{gate closed: keep the last value}) \end{cases}$$

$\beta = 0.05$ (running-average weight: smaller = smoother but slower) $\gamma = 10^{-3} \mu\text{m}$ (freeze $\hat{a}'$ once $|\Delta h_d|$ drops below this)

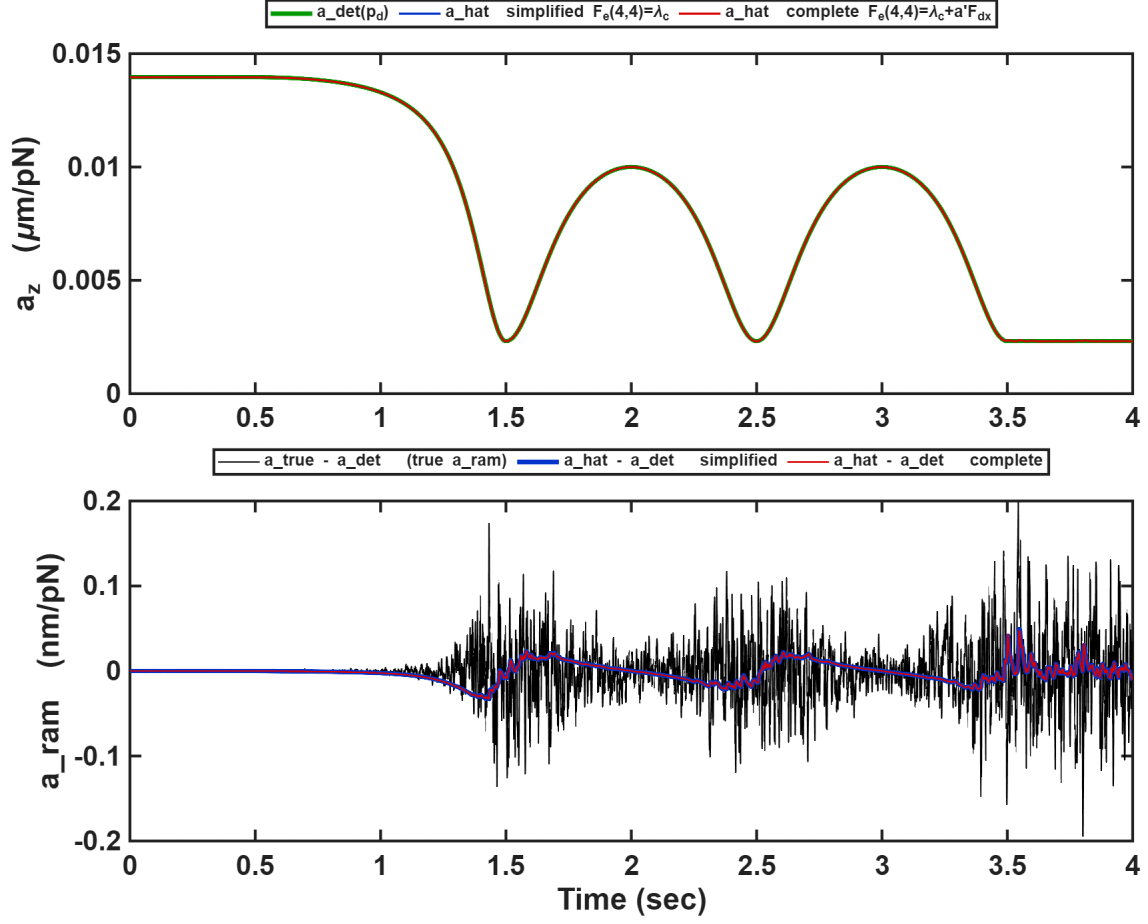
This one self-difference supplies a' everywhere in both runs; the runs differ *only* in how the gain *level* $a_{x_{det}}$ is anchored:

- **original (RW):** $a_{x_{det}} := \hat{a}_x$ (no outside anchor — the estimate leans on itself), $\mathbf{F}_e(4, 4) = 1$, $Q_{44} = \frac{2}{1 + \lambda_c} \hat{a}_x'^2 \sigma_{\delta h}^2$ — *fully* model-free.
- **AR(1):** the level $a_{x_{det}} = a_{\text{nom}}/c(\bar{h}_d)$ is still taken from the wall, $\mathbf{F}_e(4, 4) = \lambda_c$, $Q_{44} = (3 - 2\lambda_c^2) \hat{a}_x'^2 \sigma_{\delta h}^2$ — a' is model-free, but the level still needs the wall.



Same self-difference $\hat{a}'_x$ in both runs. Top: gain a_z . Bottom: slope a'_z . With a known level anchor (AR(1)) the estimate follows the truth; without one (RW) it is buried in noise.

Verification — $F_e(4,4) = \lambda_c$ (simplified) vs $\lambda_c + a'_x F_{dx}$ (complete), known $c(\bar{h})$, same seed



Top: $a_z - a_{x_{\text{det}}}(p_d)$ and the two $\hat{a}_x$. Bottom: $a_x - a_{x_{\text{det}}}$ (true, black) and $\hat{a}_x - a_{x_{\text{det}}}$ (both versions). The two versions coincide: $\max |\hat{a}_x^{\text{complete}} - \hat{a}_x^{\text{simplified}}| / a_{x_{\text{det}}} = 0.085\%$; $a_{x_{\text{det}}}(p_d) - \hat{a}_x = -0.03\% \pm 0.30\%$ ($\bar{h}_d < 2$: $\pm 0.41\%$).